Unit 2 Guided Notes — Exponents & Scientific Notation

Same base? Move the exponents. Big or tiny number? Pack it into $a \times 10^n$.

Standards: NC.8.EE.1 · NC.8.EE.3 · NC.8.EE.4

This packet weaves three parts together: **Guided Notes** with worked examples, **Practice** built into each section to try as you learn, and a **Cumulative Review** with a full **Answer Key** at the end. Fill in the blanks and show your work on your printed copy.

Topics Covered: 2.1 Exponent Rules — Product, Quotient, Power, Zero, Negative (NC.8.EE.1) · 2.2 Scientific Notation — Format & Conversions (NC.8.EE.3) · 2.2 Multiplying & Dividing Scientific Notation (NC.8.EE.4) · 2.2 Word Problems with Scientific Notation (NC.8.EE.4)

⚠ **Parentheses come first!** A power attaches only to what it touches. $-3^2 = -9$, but $(-3)^2 = 9$. And a power outside parentheses hits EVERYTHING inside: $(3x^2)^3 = 27x^6$ (the 3 is cubed too). Keep an eye on parentheses all unit long.

1 Product & Quotient Rules

↗ **KEY RULE:** Exponent rules only apply when the BASES are the SAME!

Fill the examples and key phrases on your printed copy.

Rule	Formula	Simple Example	Monomial Example	Key Phrase
Product Rule	$x^a \times x^b = x^{a+b}$	__________	__________	__________
Quotient Rule	$x^a \div x^b = x^{a-b}$	__________	__________	__________

PRODUCT RULE: COUNT THE FACTORS

x x x × x x = x x x x x

x^3 (3 factors) x^2 (2) x^5 (5 factors)

$x^3 \times x^2 = x^{3+2} = x^5$ → ADD the exponents

See WHY

Product Rule · $x^3 \times x^2 = x^5$

See why → x x x × x x = x x x x x = x^5

Count the factors: 3 + 2 = 5. Same base → ADD the exponents.

Quotient Rule · $x^5 \div x^2 = x^3$

See why → x x x x x ÷ x x → cancel 2 pairs → x x x = x^3

Cancel matching factors: 5 − 2 = 3. Same base → SUBTRACT the exponents.

Quick Check

MC **Simplify $x^5 \cdot x^3$. Which rule applies, and what is the result?**

◯ Product Rule → x^8 ◯ Power Rule → x^{15} ◯ Quotient Rule → x^2

Worked Example — Simplify $(4x^3y^2)(3x^5y)$

Step 1 — Multiply coefficients: $4 \times 3 = 12$

Step 2 — Add x exponents: $x^3 \cdot x^5 = x^8$

Step 3 — Add y exponents: $y^2 \cdot y^1 = y^3$

✓ **Step 4 — Combine:** $12x^8y^3$

PRACTICE 1 Product & Quotient

Rules: Multiply → ADD exponents · Divide → SUBTRACT. Write each quotient as a reduced fraction. **Show your work in the box.**

1. Simplify $x^4 \cdot x^7$

2. Simplify $(3x^2)(5x^6)$

3. Simplify $(2a^3b^2)(4a^2b^5)$

4. Simplify $(-3x^4y)(6xy^3)$

5. Simplify $(5m^2n)(2mn^3)(3n^2)$

6. Simplify $\frac{x^9}{x^4}$

7. Simplify $\frac{15a^7}{3a^2}$

8. Simplify $\frac{24x^5y^6}{6x^2y^2}$

9. Simplify $\frac{20m^8n^3}{5m^3n}$

10. Simplify $\frac{-18p^6q^5}{3p^2q^2}$

11 Simplify $\frac{(2x^3)(6x^4)}{4x^2}$

12 Simplify $\frac{(3a^2b)(4a^3b^2)}{6ab}$

2 Power Rules

A power outside parentheses applies to every factor inside. Fill in, then check.

Rule	Formula	Simple Example	Monomial Example	Key Phrase
Power Rule	$(x^a)^b = x^{a \times b}$	________	________	________
Power of a Product	$(xy)^n = x^n y^n$	________	________	________
Power of a Quotient	$\left(\frac{x}{y}\right)^n = \frac{x^n}{y^n}$	________	________	________

⚠ PARENTHESES MATTER!
$-3^2 =$ ________ vs. $(-3)^2 =$ ________
Without ( ), ONLY the base gets the exponent — not the negative sign!

See WHY

Power Rule · $(x^2)^3 = x^6$

See why → $x^2 \; x^2 \; x^2 = x \; x \cdot x \; x \cdot x \; x = x^6$

Three copies of x^2: $2 \times 3 = 6$. Power to a power → MULTIPLY.

ERROR ANALYSIS **Spot the Error — Simplify $(3x^2)^3$.** A student wrote $3x^6$.
The coefficient 3 must ALSO be cubed: $3^3 = 27$. Correct: $\mathbf{27x^6}$. The power rule applies to EVERYTHING inside the parentheses.

Worked Example — Simplify $(2a^3b)^4$

Step 1 — Power the coefficient: $2^4 = 16$

Step 2 — Multiply the a exponent: $a^{3 \times 4} = a^{12}$

Step 3 — Multiply the b exponent: $b^{1 \times 4} = b^4$

✓ **Step 4 — Combine:** $16a^{12}b^4$

PRACTICE 2 **Power Rules**

Rule: Power to a power → MULTIPLY; apply the power to EVERY factor, top and bottom.

1

Simplify $(x^4)^5$

2

Simplify $(2x^3)^4$

3

Simplify $(-3a^2)^3$

4

Simplify $(x^2y^5)^3$

5

Simplify $(4m^3n^2)^2$

6

Simplify $(2x^3y^4)^5$

7

Simplify $\left(\frac{x^2}{y^3}\right)^4$

8

Simplify $\left(\frac{2a}{b^2}\right)^3$

9

Simplify $\left(\frac{3x^4}{2y}\right)^2$

10

Simplify $((x^3)^2)^4$

11

Simplify $(2x^2y)^3(3xy^2)^2$

12

Simplify $(-2a^3b^2)^4$

3 Zero & Negative Exponents

Anything to the 0 power is 1; a negative exponent flips to a fraction.

Rule	Formula	Simple Example	Monomial Example	Key Phrase
Zero Exponent	$x^0 = 1$	________	________	________
Negative Exponent	$x^{-n} = \frac{1}{x^n}$	________	________	________

See WHY

Zero Exponent · $x^0 = 1$

See why → x x x ÷ x x x = 1, but by the quotient rule = $x^{3-3} = x^0$. So $\boxed{x^0 = 1}$

Anything (except 0) divided by itself is 1 → $x^0 = 1$.

Negative Exponent · $x^{-3} = \frac{1}{x^3}$

See why → x x ÷ x x x x x = $x^{2-5} = x^{-3}$; cancelling leaves $\boxed{\frac{1}{x^3}}$

A negative exponent means 'flip it' → $x^{-n} = \frac{1}{x^n}$.

Match the Rule to its Key Phrase

MATCH Draw a line from each rule to its key phrase.

Rule	Key Phrase
Product Rule	Power to power? MULTIPLY
Quotient Rule	Same base? ADD exponents
Power Rule	Flip to a fraction
Negative Exponent	Same base? SUBTRACT exponents

Negative Exponents Across the Fraction Bar

A negative exponent moves its factor to the OTHER side of the bar and turns positive.

Move across the bar: $x^{-3} = \frac{1}{x^3}$ · $\frac{1}{x^{-3}} = x^3$ · $\frac{x^{-2}}{y^{-4}} = \frac{y^4}{x^2}$

In parentheses — flip to the reciprocal: $(\frac{a}{b})^{-n} = (\frac{b}{a})^n$. $(\frac{2}{3})^{-2} = (\frac{3}{2})^2 = \frac{9}{4}$ · $(\frac{x^2}{y})^{-3} = (\frac{y}{x^2})^3 = \frac{y^3}{x^6}$

PRACTICE 3 Zero & Negative Exponents

Write every answer with POSITIVE exponents only, as a reduced fraction.

1 Simplify $(7x^4y^2)^0$

2 Simplify $(3a^2b)^0$

3 Evaluate 4^{-3}

4 Positive exponents: $2x^{-5}$

5

Simplify $\frac{x^{-3}}{y^{-2}}$

6

Simplify $\frac{a^{-2}b^{3}}{c^{-1}}$

7

Simplify $\frac{12m^{-2}n^{4}}{3m^{4}n^{-1}}$

8

Simplify $\frac{8x^{-3}y^{2}}{2xy^{-1}}$

9

Simplify $(\frac{x}{y})^{-2}$

10

Evaluate $(\frac{2}{3})^{-3}$

11

Simplify $(\frac{a^{2}}{b})^{-3}$

12

Simplify $(\frac{3x}{y^{2}})^{-2}$

2 Scientific Notation

$$a \times 10^n \quad \text{where} \quad 1 \le a < 10$$

4.5×10^7 | coefficient · base (always 10) · exponent

SCIENTIFIC NOTATION: MOVE THE DECIMAL

$$45{,}000 \rightarrow 4.5 \times 10^4$$

big number → move LEFT 4 places → exponent +4

$$0.0067 \rightarrow 6.7 \times 10^{-3}$$

small number → move RIGHT 3 places → exponent −3

Type	Direction	Exponent	Example
Large number	Move decimal LEFT	______	$45{,}000 = 4.5 \times 10^4$
Small number (< 1)	Move decimal RIGHT	______	$0.0067 = 6.7 \times 10^{-3}$

PRACTICE B Scientific Notation Conversions

Remember: $1 \le a < 10$. Positive exp = BIG · Negative exp = SMALL.

1. 45,000,000 → sci. notation

2. 0.00032 → sci. notation

3. 6.02×10^8 → standard

4. 9.1×10^{-5} → standard

5. 780,000 → sci. notation

6. 0.0000067 → sci. notation

7. 3.45×10^4 → standard

8. 1.2×10^{-3} → standard

9 $5{,}280{,}000{,}000$ → sci. notation

10 2.5×10^{-2} → standard

11 0.000000904 → sci. notation

12 7.15×10^{6} → standard

13 $63{,}400{,}000{,}000$ → sci. notation

14 8.8×10^{-6} → standard

Multiplying Scientific Notation — 3-Step Process

1. ___________ the coefficients (the a-values)
2. ___________ the exponents
3. Adjust if coefficient is NOT between _______ and _______

Example (no adjustment): $(3 \times 10^4) \times (2 \times 10^5)$

Multiply coefficients: $3 \times 2 = 6$

ADD exponents: $4 + 5 = 9$

✓ **Answer:** 6×10^9

Dividing Scientific Notation — 3-Step Process

1. ___________ the coefficients
2. ___________ the exponents
3. Adjust if needed

Example (no adjustment): $(8 \times 10^6) \div (4 \times 10^2)$

Divide coefficients: $8 \div 4 = 2$

SUBTRACT exponents: $6 - 2 = 4$

✓ **Answer:** 2×10^4

⚠ **ADJUSTMENT RULES:**
- If coefficient ≥ 10: Move decimal LEFT, ADD 1 to exponent
- If coefficient < 1: Move decimal RIGHT, SUBTRACT 1 from exponent

Example needing adjustment: $(5 \times 10^3) \times (4 \times 10^2)$

Multiply: $5 \times 4 = 20$

Add exponents: $3 + 2 = 5 \rightarrow$ result: 20×10^5

Adjust ($20 \geq 10$): $20 = 2.0 \times 10^1$, so ADD 1 to exponent

✓ **Final answer:** 2.0×10^6

ERROR ANALYSIS **Spot the Error — Convert 0.0045 to scientific notation.** A student wrote 4.5×10^3. Small number → NEGATIVE exponent. Correct: $\mathbf{4.5 \times 10^{-3}}$. Numbers less than 1 always have negative exponents in scientific notation.

PRACTICE C Operations with Scientific Notation

Multiply/divide coefficients · add/subtract exponents · adjust so $1 \leq a < 10$.

1. $(3 \times 10^4)(2 \times 10^5)$

2. $(8 \times 10^7) \div (4 \times 10^3)$

3. $(5 \times 10^3)(7 \times 10^6)$

4. $(9.6 \times 10^8) \div (3.2 \times 10^5)$

5. $(6 \times 10^5)(5 \times 10^4)$

6. $(1.5 \times 10^{-3})(4 \times 10^7)$

7. $(7.2 \times 10^9) \div (1.8 \times 10^{-2})$

8. $(2.5 \times 10^6)(8 \times 10^{-3})$

Word Problems with Scientific Notation

Strategy: Identify what operation to use. Write your scientific-notation setup. Adjust if needed.

PRACTICE D Word Problems

Identify the operation, write your setup, then adjust so $1 \leq a < 10$.

1. The Sun is about 9.3×10^7 miles from Earth. Light travels 1.86×10^5 miles per second. How many seconds for light to reach Earth?

Answer: ____________________

2. A bacteria cell is about 2×10^{-6} m long; a virus about 5×10^{-8} m. How many times longer is the bacteria?

Answer: ____________________

3. The US debt is about 3.4×10^{13} dollars; population about 3.4×10^8. What is the debt per person?

Answer: ____________________

4. A grain of sand has mass 6.5×10^{-4} g. A beach has 8×10^{18} grains. Total mass in grams?

Answer: ____________________

5. A body has 3.7×10^{13} cells, each 1×10^{-5} m wide. Lined up, how long in meters?

Answer: ____________________

6. A computer does 4.8×10^9 calculations per second. A task needs 1.44×10^{12}. How many seconds?

Answer: ____________________

7. A hard drive holds 2×10^{12} bytes. Each photo is 4×10^6 bytes. How many photos fit?

Answer: ____________________

8. Light travels 3×10^8 m/s. How far in 6×10^3 seconds?

Answer: ______________________________

★ Cumulative Practice — Unit 2 Review ★

Mixed practice over everything above. Show all work; circle your final answer.

REVIEW · A Exponent Rules

1. Simplify $x^4 \cdot x^6$

2. Simplify $(3x^2)(4x^5)$

3. Simplify $(2a^3b^2)(5ab^4)$

4. Simplify $\frac{24x^7}{6x^3}$

5. Simplify $(m^3)^5$

6. Simplify $(2x^2y^3)^4$

7. Simplify $\frac{18a^5b^6}{3a^2b^2}$

8. Simplify $(\frac{x^4}{y^2})^3$

REVIEW · B Scientific Notation Conversions

1. 7,200,000 → sci. notation

2. 0.00045 → sci. notation

3. 3.05×10^7 → standard

4. 6.1×10^{-5} → standard

5

$91{,}000 \rightarrow$ sci. notation

6

$0.0000008 \rightarrow$ sci. notation

7

$4.44 \times 10^6 \rightarrow$ standard

8

$2.7 \times 10^{-3} \rightarrow$ standard

REVIEW · C Operations

1

$(3 \times 10^4)(3 \times 10^5)$

2

$(8 \times 10^9) \div (2 \times 10^3)$

3

$(5 \times 10^6)(4 \times 10^3)$

4

$(9.6 \times 10^8) \div (1.6 \times 10^2)$

5

$(7 \times 10^5)(8 \times 10^2)$

6

$(1.2 \times 10^{-3})(4 \times 10^5)$

REVIEW · D Word Problems

1. A red blood cell is about 8×10^{-6} m wide; a platelet about 2×10^{-6} m. How many times wider is the red blood cell?

Answer: ______________________

2. Earth is about 1.5×10^8 km from the Sun. A spacecraft travels 3×10^4 km/h. How many hours for the trip?

Answer: ______________________

3. A factory makes 5×10^3 widgets per day. How many in 2×10^2 days?

Answer: ____________________

4. A virus is about 3×10^{-7} m and a cell about 6×10^{-5} m. How many times larger is the cell?

Answer: ____________________

REVIEW · E Negative Exponents & Rational Expressions

Write every answer with positive exponents only, as a reduced fraction.

1 Evaluate 5^{-2}

2 Positive exponents: $3x^{-4}$

3 Simplify $\frac{x^{-5}}{y^{-3}}$

4 Simplify $(\frac{a}{b})^{-3}$

5 Simplify $\frac{10a^{-2}b^5}{2a^3b^{-1}}$

6 Simplify $(\frac{2x^2}{y})^{-2}$